

Yibin (Leon) Liu

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Education

NUS Computing, National University of Singapore

Jan 2027 – June 2031

Incoming Ph.D. in Computer Science and Robotics

- **Advisor:** Prof. Weiyu Liu[✉]

Northeastern University, Shenyang, China[✉]

Sept 2022 – June 2026

Bachelor of Artificial Intelligence

- **Achievements:** 88/100 (Overall), 93/100 (Junior)
- **Programs:** 985, 211, Double First Class University

Global Innovation Exchange - UW Seattle & Tsinghua[✉]

Jul 2024 – Oct 2024

Access Computing Summer Program, AI & HCI

A collaboration between the University of Washington, Tsinghua University, and Microsoft, focused on AI and HCI innovation. Full scholarship awarded.

About

My research lies at the intersection of Language Grounding, Multimodal Reasoning and Planning, and Human-Robot Interaction. I am particularly interested in developing generalizable mobile manipulation embodied agents that ground, plan, and act in open-world environments, enabling seamless collaboration and interaction with humans. I possess **full-stack** research and engineering capabilities in embodied AI for mobile manipulation, **spanning from building simulation, model training, to real-world robot deployment and benchmark optimization.**

Publications

From Passive Observer to Active Critic: Reinforcement Learning Elicits Process Reasoning for Robotic Manipulation[✉]

ECCV 2026 (Under review, Top 25%)

Y Liu*, Y Lyu*, D Gao*, Z Liang, W Tang, S Mu, X Yang, Y Mu

RoboTwin 2.0: A Scalable Data Generator and Benchmark with Strong Domain Randomization for Robust Bimanual Robotic Manipulation[✉]

ICML 2026 (2.4k+ GitHub Stars)

Co-first author

Tianxing Chen*, Zanxin Chen*, Baijun Chen*, Zijian Cai*, **Yibin Liu***, ... Ping Luo†, Yao Mu†

Widely adopted bimanual manipulation benchmark with 5 embodiments, 50 dual-arm tasks, and 700+ assets.

Vlaser: Vision-Language-Action Model with Synergistic Embodied Reasoning[✉]

ICLR 2026

Co-author

Ganlin Yang, Tianyi Zhang, Haoran Hao, Weiyun Wang, **Yibin Liu**, ... Jifeng Dai, Jiangmiao Pang, Gen Luo, Wenhai Wang, Yao Mu†, Zhi Hou†

Proposes a unified VLA model enabling synergistic embodied reasoning across perception, planning, and control.

The Homework Wars: Exploring Emotions, Behaviours, and Conflicts in Parent-Child Homework Interactions[✉]

*ACM IMMUT/Ubicomp 2025, **Advisor First-Author***

Nan Gao*, **Yibin Liu**, Xin Tang, Yanyan Liu, Chun Yu, Yun Huang, Xuhai Xu, Jun Wei, Yuanchun Shi

First work to leverage LLM-human collaboration for qualitative analysis to infer human behavioural states.

HyCodePolicy: Hybrid Language Controllers for Multimodal Monitoring and Decision in Embodied Agents[✉]

ICCV 2025 Workshop on Multi-Modal Reasoning for Agentic Intelligence

Yibin Liu*, Zhixuan Liang*, Zanxing Chen*, Tianxing Chen, ... , Yuseng Qing, Yao Mu

Self-Guide: A LLM Reasoning Enhancement Method Based on Self-Guided Planning.[✉]

Journal of Chinese Information Processing (2024)

Yibin Liu, Zhenghao Liu, Yukun Yan, Shi Yu, Shuo Wang, Liner Yang, Yu Gu, Ge Yu, Huimin Chen

Revealed that enhancing LLM agent reasoning relies on injecting useful information rather than debate/reflection.

Research Experience

University of North Carolina at Chapel Hill – Research Intern

Jun 2025 – Mar 2026, Remote

Advisor: Prof. Mingyu Ding[✉]

- Enhancing MLLMs with spatial understanding and action-level reasoning for robotic manipulation via human language instructions and AR-guided interactions.

Shanghai Jiao Tong University – Research Assistant

Mar 2025 – Jan 2026, Shanghai, China

Advisor: Prof. Yao Mu[✉]

- Core contributor to RoboTwin 2.0, leading robot policy code generation for scalable bimanual manipulation.

Pervasive HCI Lab, Tsinghua University[✉] – Research Assistant

Jun 2024 – Jan 2025, Beijing, China

Advisor: Prof. Nan Gao[✉], Prof. Chun Yu[✉]

- Conducted research in using LLM and HCI to infer human behaviors and mental states. Our research represents an application of LLMs to encode and analyze human behavior through dialogue data directly.

Industry Experiences

Robbyant[✉] (part of Ant Group[✉]) – Research Intern

Mar 2026 – Present, Shanghai, China

- **Focus:** Large-scale Foundation Models for mobile manipulation, including VLA pre-training, post-training, real-robot deployment, and automated benchmark verification.

Horizon Robotics – Cloud Platform Intern

Jun 2025 – Jan 2026, Beijing, China · Hybrid

Mentor: Yusen Qin (VP of Technology, D-Robotics)[✉]

- Developing the RDK-agent, building an LLM-powered Copilot for robotics development.

Academic Activities

- **Community:** Co-Founder of VapourX[✉], an open community for embodied AI beginners, and researchers.
- **Workshop Service:** Student Committee of TriFusion Workshop @ SIGGRAPH Asia 2025[✉]
- **Academic Service:** Reviewer for CHI 2026, Chinese CHI 2024.

Awards

- 2025.07 Outstanding Poster at ChinaSI 2025 (**Ranking 1st among 61 posters**, RoboTwin 2.0).
- 2024.11 Outstanding Individual in Technological Innovation of Northeastern University.
- 2024.05 **Finalist** in Mathematical Contest in Modeling (MCM/ICM 2024, **Top 1.69% of 10,387 teams**).
- 2023.10 National Level Third Prize in **RoboCup** China Competition, Simulation 3D League.
- 2023.10 National Level Second Prize in **FIRA SimuroSot** China Competition.
- 2023.11 Future Technology Taihu Scholarship & Excellent Student Scholarship at Northeastern University.

Projects

RDK Copilot: LLM-powered Copilot for Horizon Robotics

Mar – May 2025

- Developed and deployed a VSCode plugin that assists robotic system development using LLMs. Supported features include automatic coding, environment setup, code completion, and manipulation data acquisition.

MinRL: Minimal, Clean Code for Reinforcement Learning (114 Stars)

GitHub Repository[✉]

- **Recognized and pinned by MathFoundationRL[✉], the most popular RL course on Chinese platforms** Developed a RL framework featuring clean implementations of fundamental algorithms.

Bencao RAG Medical Intelligent Assistant

GitHub Repository[✉]

- Developed a medical knowledge question-answering system that integrates context awareness, internet access, knowledge graphs, and RAG method to provide accurate and personalized medical information.